Penetration Testing & Cryptography in a Virtual Lab

Takwa Sassi

This project aims to:

- Simulate real-world attacks using penetration testing tools
- Analyze system weakness
- Apply cryptographic and decryption techniques to secure the system
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1. Tools & Environment
    - Oracle VM VirtualBox: https://www.virtualbox.org/wiki/Downloads
    - Kali Linux (attacker machine): https://www.kali.org/get-kali/
    - Metasploitable2 (vulnerable machine) : https://sourceforge.net/projects/metasploitable2/
    - Metasploit Framework (MSF)
    - Wireshark (Network analysis)

2. System Configuration
    - Kali Linux (attacker):
      Operating system: Debian
      Base memory (RAM): 4096MB
      Processor: 2core
      SATA Port0 : kali_linux.vdi (25 GB)
      Network: Host-only Adapter

    - Metasploitable2(target):
      Operating system: Ubuntu
      Base memory: 512MB
      SATA Port0 : Metasploitable.vmdk (8GB)
      Network: Host-only Adapter

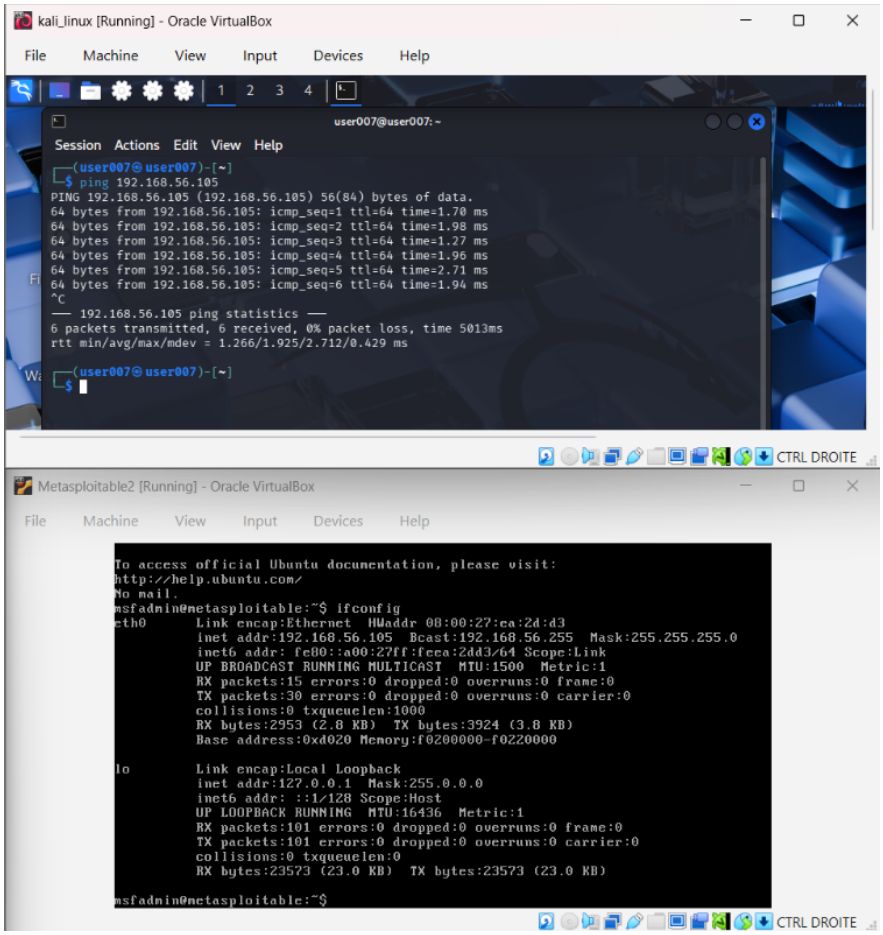
    ✓ Both machines must be on **Host-Only Adapter**; it proves that both machines are in the same network, and we can simulate a real attack environment safely.

    Target information (Metasploitable2)

    IP Adress: 192.168.56.105

3. Attack Phases

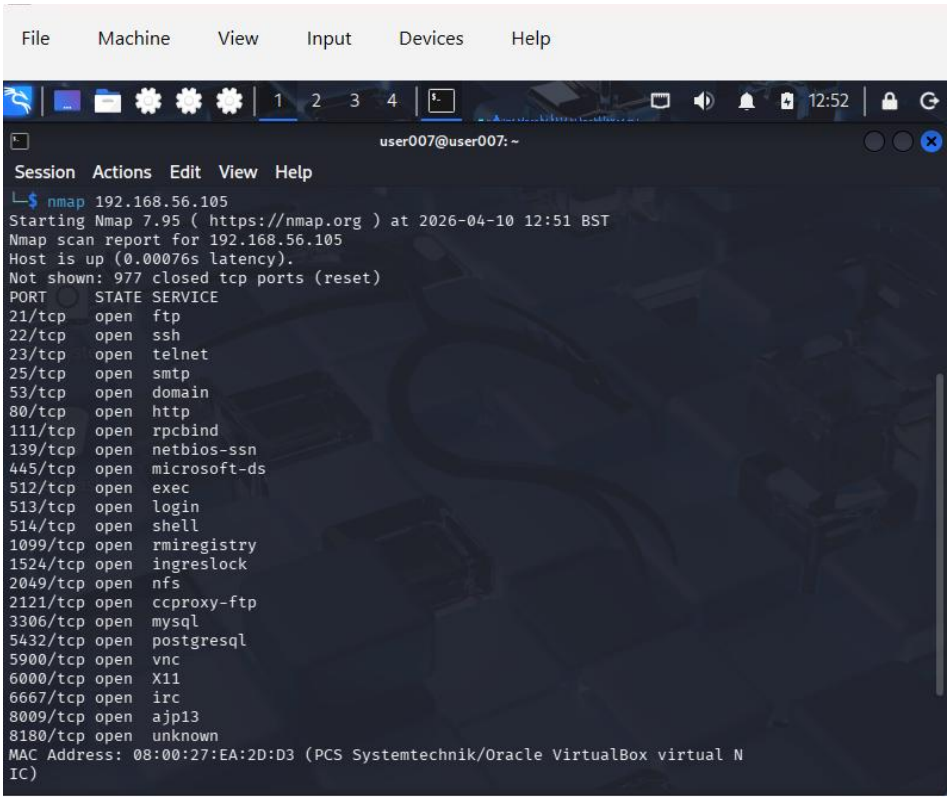
a) Connectivity test



- Connectivity between attacker and target machines was successfully established.

b) Scanning phase

- Network scanning



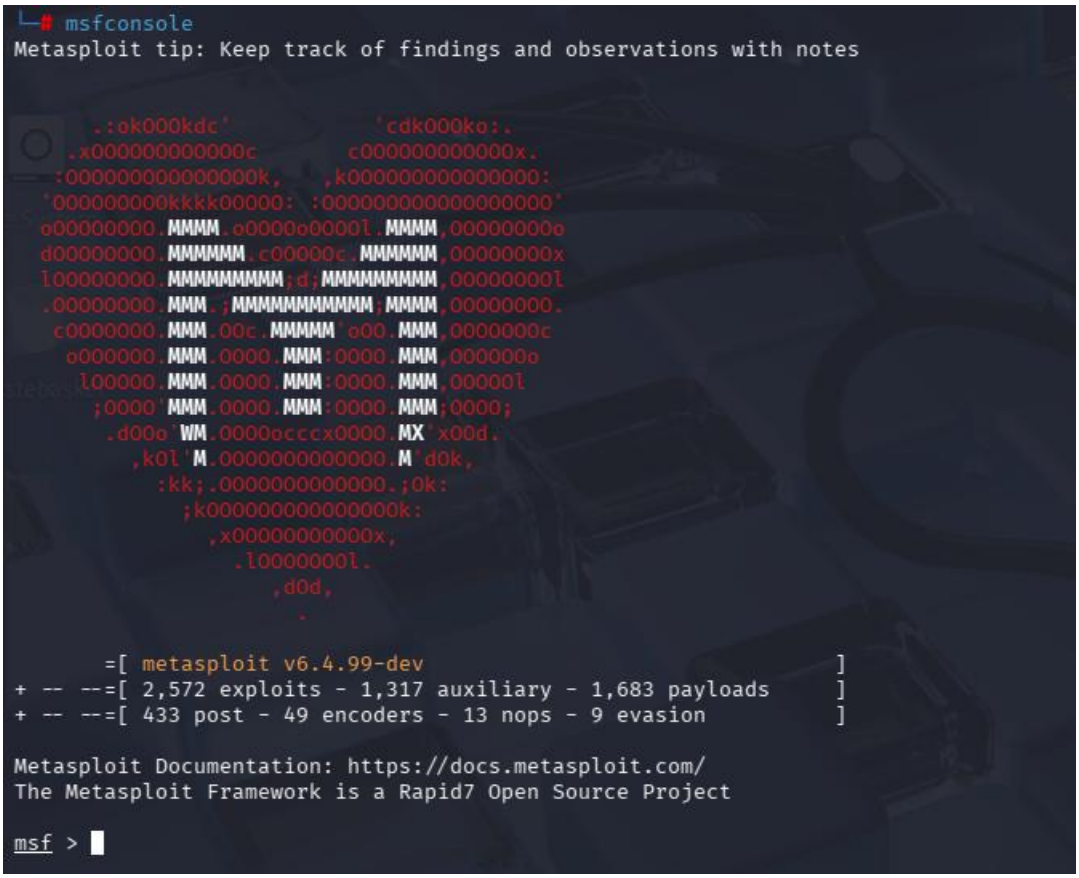
- ➢ Multiple services are exposed(open), vulnerable.
- ➢ Communication is not encrypted

c) Exploitation phase(attack)

- Get into root
- Initialize Metasploit database



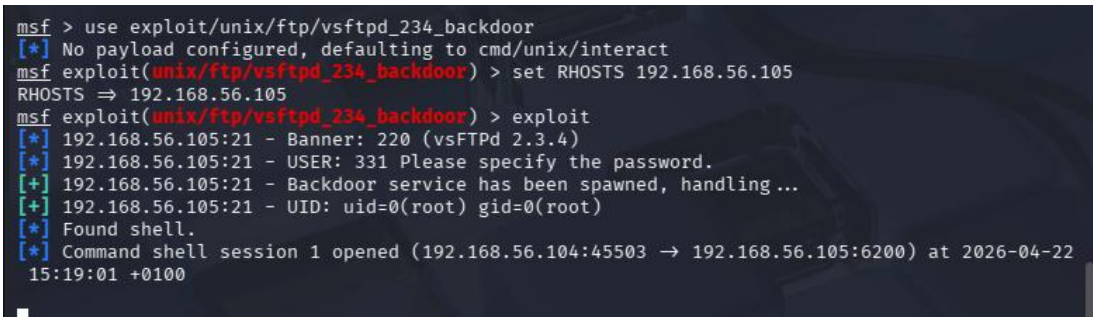
- Launch Metasploit



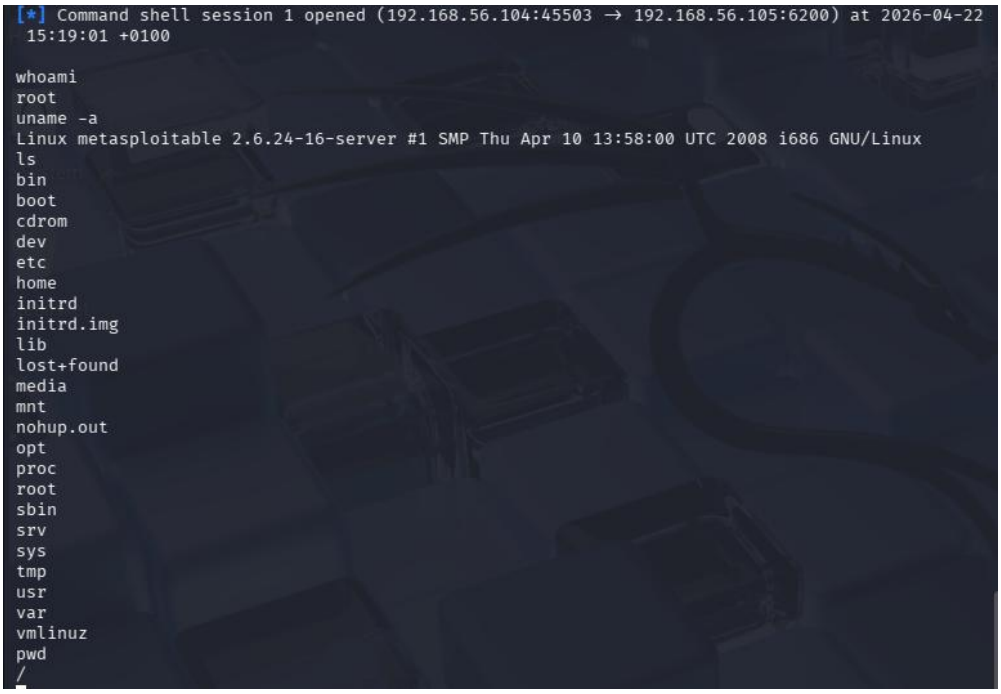
- Search vulnerabilities



- Manually check module path, then set target (victim Ip)



- Results



➢ The system is vulnerable.

4. Defense phase
    - Password Hashing (Implementation python)

```
┌──(user007㉿user007)-[~]
└─$ nano hash_demo.py
```

```
 GNU nano 8.6                                hash_demo.py *
import hashlib
password = "msfadmin"
salt = "random_salt"

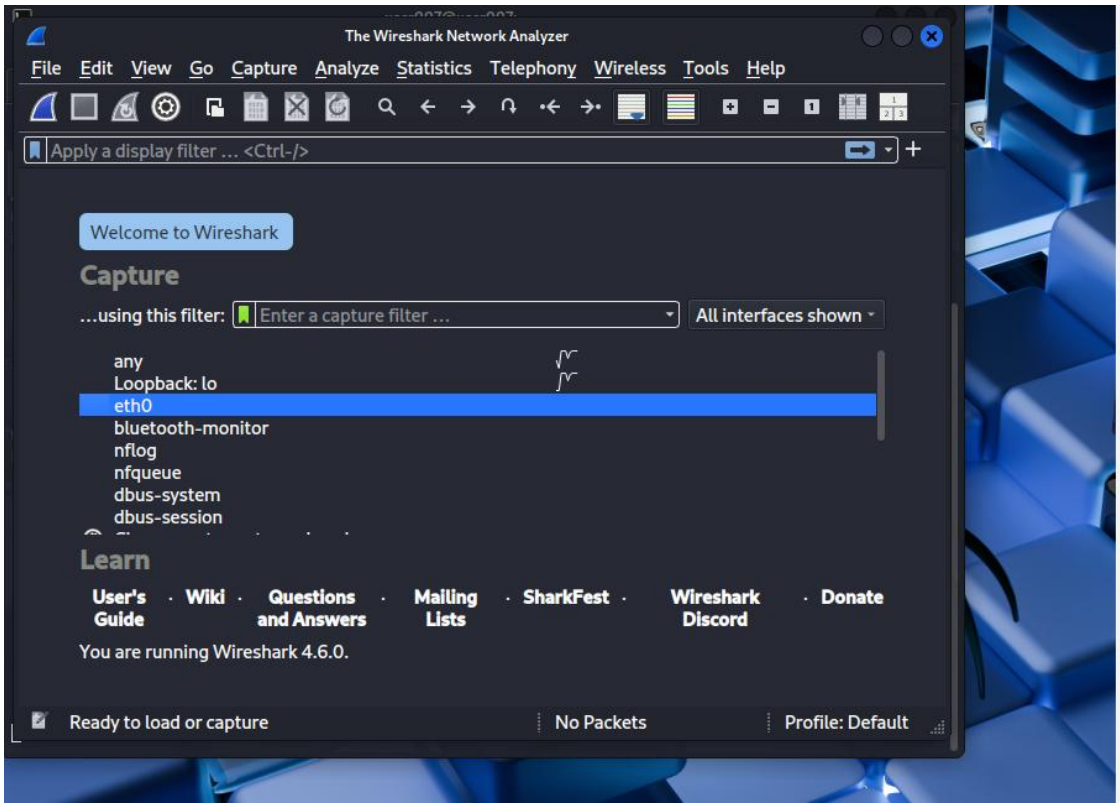
hashed = hashlib.sha256((password+salt).encode()).hexdigest()
print("Hashed password:", hashed)

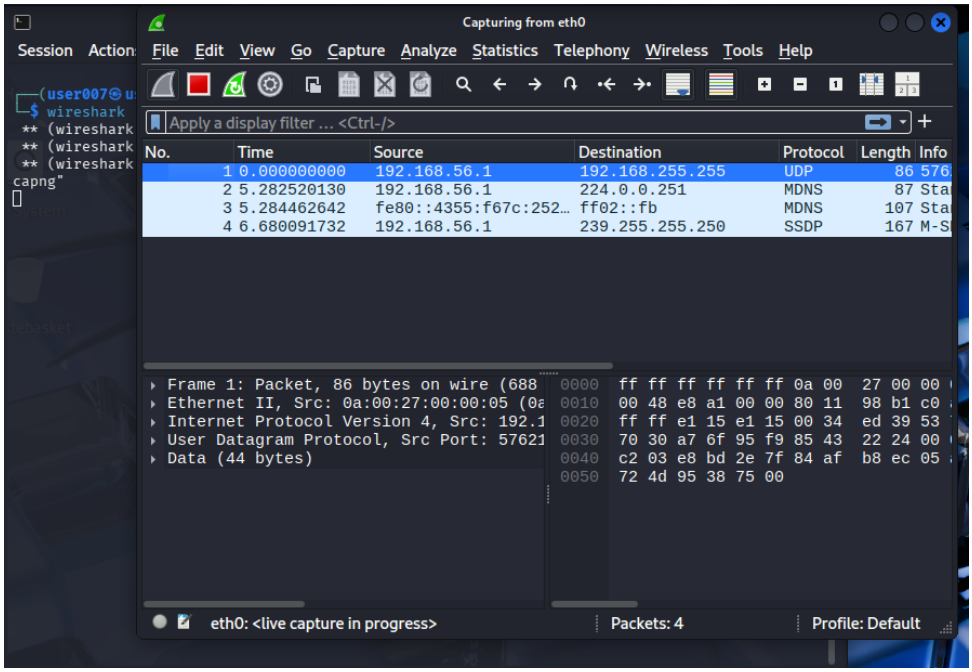
^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

- Password is transformed into a secured hash
- Protects against password leakage

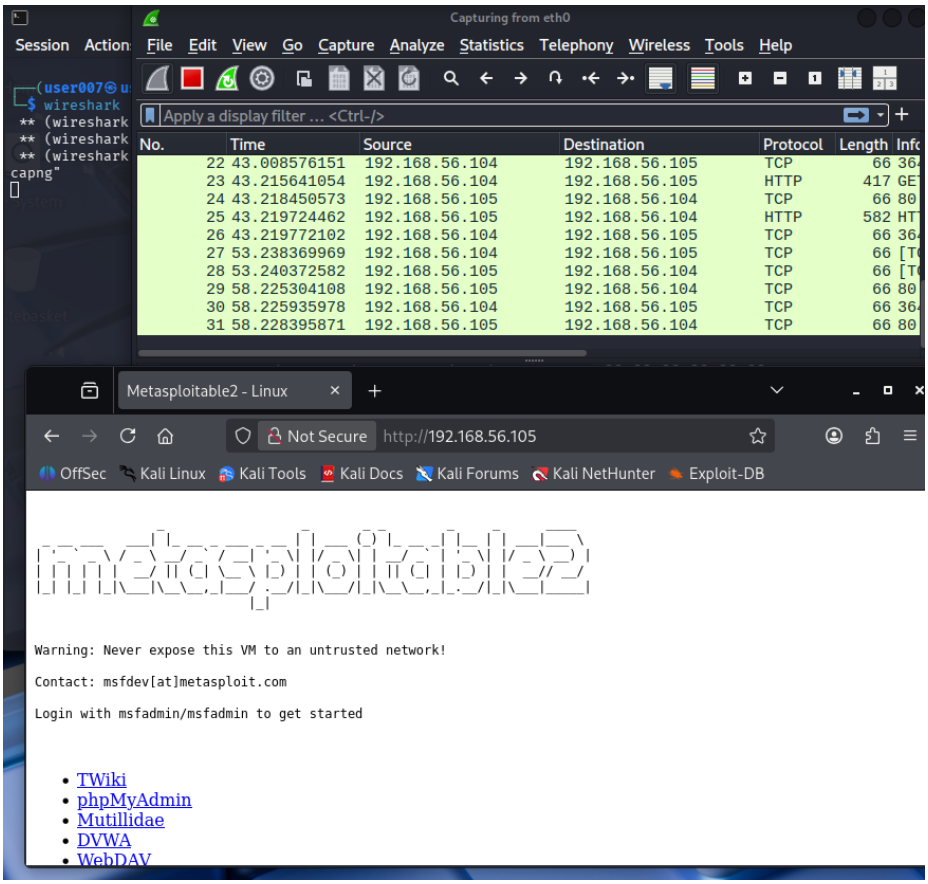
```
┌──(user007㉿user007)-[~]
└─$ python3 hash_demo.py
Hashed password: 98bc14a6403bc8093bb319c1a7770fe28c6b4aca0e83046b112014c48bfaf3c8
```

- Network traffic analysis (Wireshark)

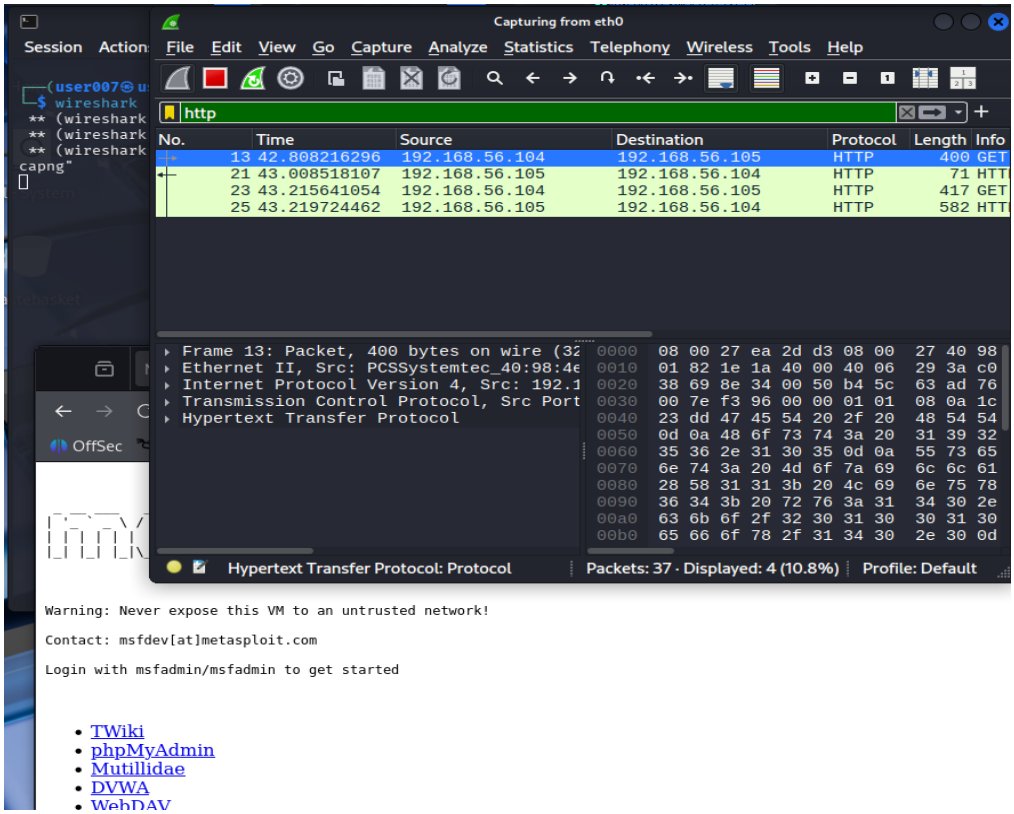




Wireshark was launched on the Kali Linux to capture live data



- Captured packets between machines
- Observing unencrypted data
- Attackers can read sensitive informations



5. Encryption & Decryption

a) Prepare the Environment (Kali)

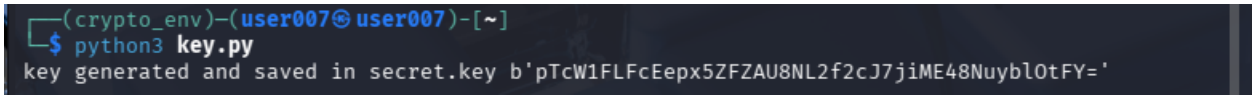
```
┌──(user007㉿user007)-[~]
└─$ python3 -m venv crypto_env

┌──(user007㉿user007)-[~]
└─$ source crypto_env/bin/activate

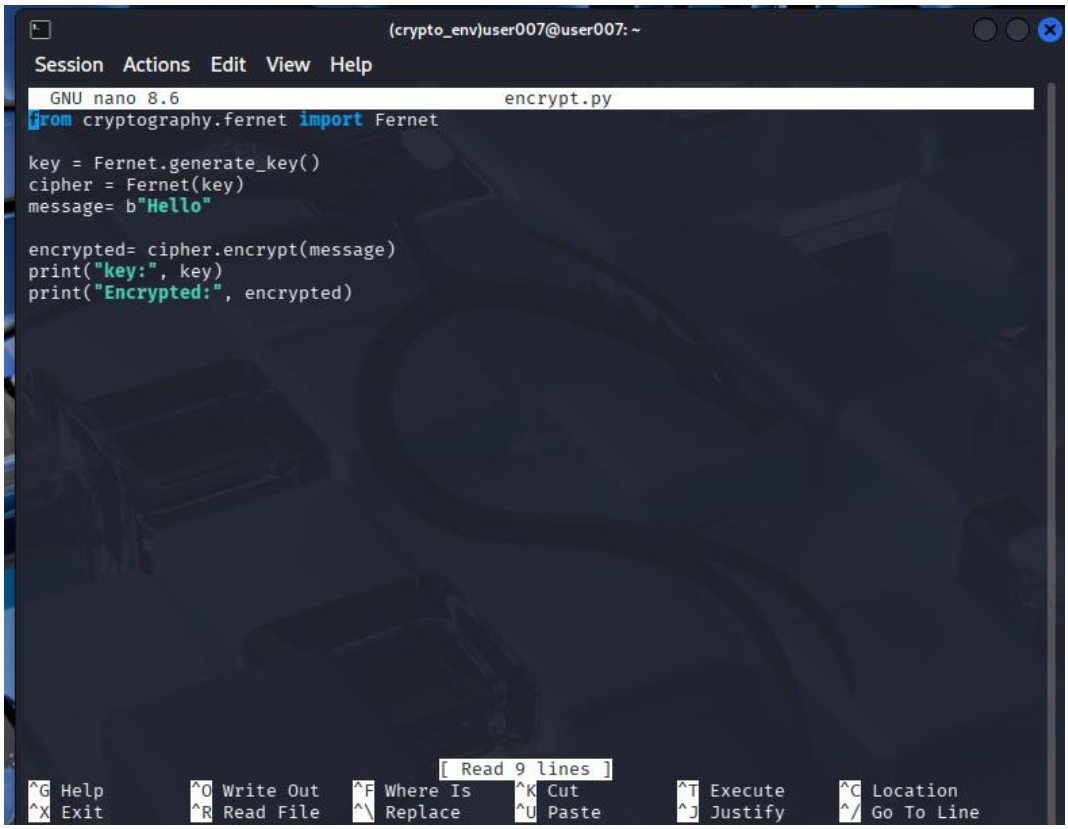
┌──(crypto_env)─(user007㉿user007)-[~]
└─$ pip install cryptography
Collecting cryptography
  Downloading cryptography-47.0.0-cp311-abi3-manylinux_2_34_x86_64.whl.metadata (4.5 kB)
Collecting cffi ⩾ 2.0.0 (from cryptography)
  Downloading cffi-2.0.0-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata
(2.6 kB)
Collecting pycparser (from cffi ⩾ 2.0.0→cryptography)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Downloading cryptography-47.0.0-cp311-abi3-manylinux_2_34_x86_64.whl (4.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.7/4.7 MB 1.1 MB/s  0:00:04
Downloading cffi-2.0.0-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (219 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Installing collected packages: pycparser, cffi, cryptography
Successfully installed cffi-2.0.0 cryptography-47.0.0 pycparser-3.0
```

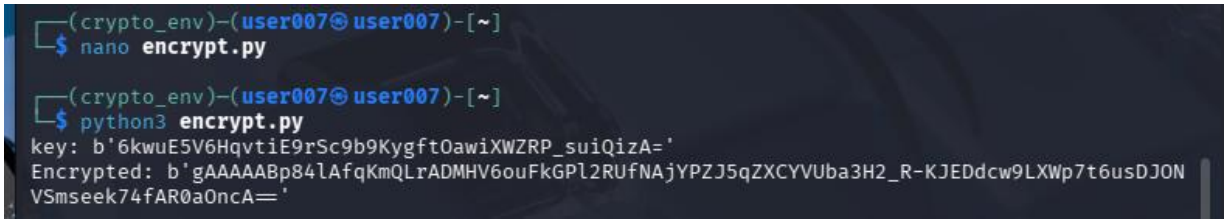
b) Generate & Store secret key



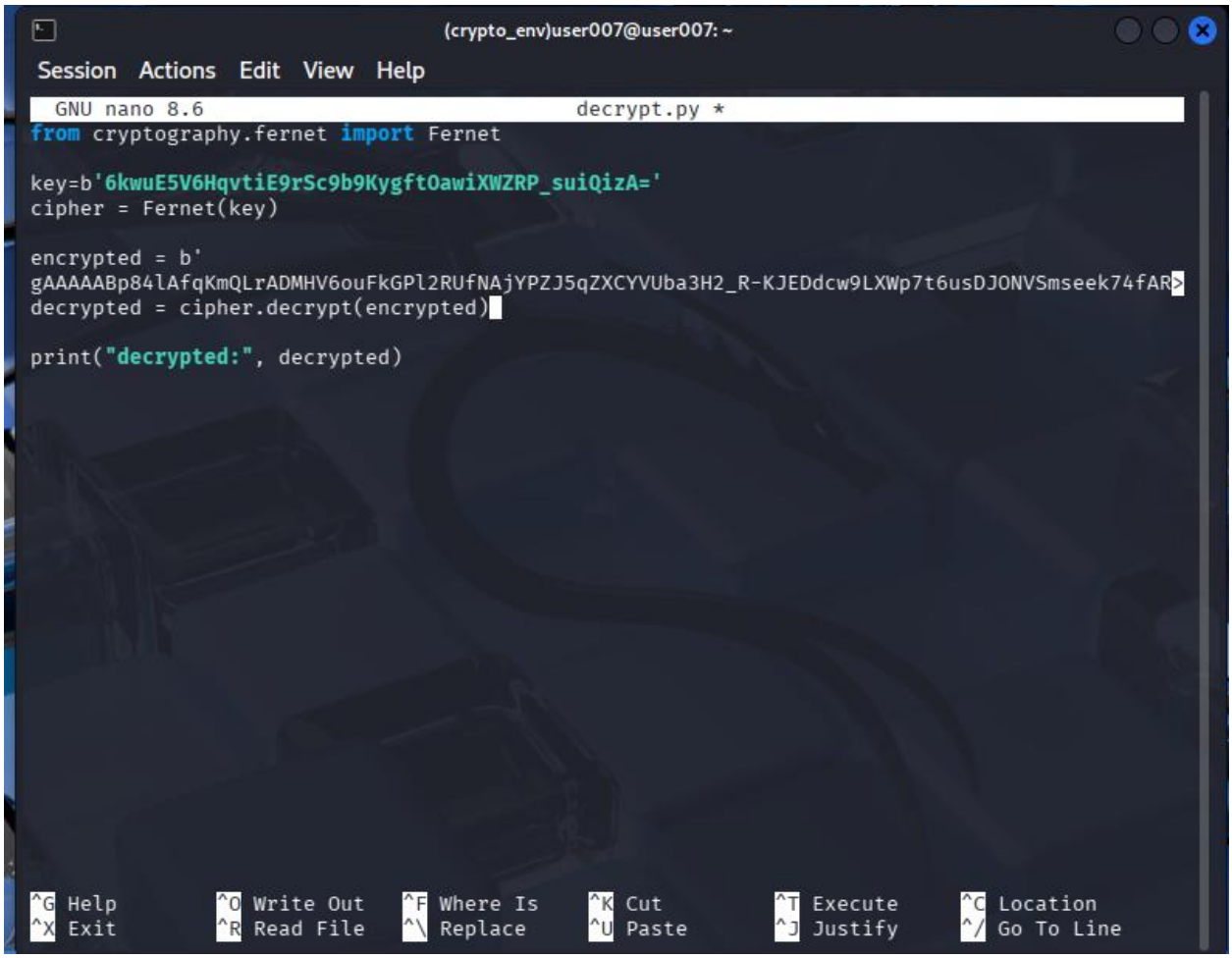


c) Encrypt a message





d) Decrypt the message





Steps	What I did	Result
Setup Lab	Installed Kali Linux and Metasploitable2	Created a safe place to test attacks
Network Setup	Connected both machines in same network	Made them communicate like real systems
Test Connection	Checked if machines can reach each other	Confirmed everything works correctly
Scan System	Scanned target for open ports and services	Found many vulnerabilities
Exploit System	Used framework to attack and get access	Saw how hackers can take control of a system
Analyze traffic	Used Wireshark to capture Network data	Discovered data was not encrypted (easy to read)
Add security	Used password hashing	Protected sensitive information
Encrypt data	Encrypted and decrypted messages	Ensured data is safe and private
Final result	Combined attack and defense	Understood that security needs both testing and protection